Semi-Supervised Locality Preserving Dense Graph Neural Network With ARMA Filters and Context-Aware Learning for Hyperspectral Image Classification

Yao Ding, Xiaofeng Zhao, Zhili Zhang, Wei Cai, Nengjun Yang, and Ying Zhan

***Abstract*— The application of graph convolutional networks (GCNs) to hyperspectral image (HSI) classification is a heavily researched topic. However, GCNs are based on spectral filters, which are computationally costly and fail to suppress noise effectively. In addition, the current GCN-based methods are prone to oversmoothing (the representation of each node tends to be congruent) problems. To circumvent these problems, a novel semi-supervised locality-preserving dense graph neural network (GNN) with autoregressive moving average (ARMA) filters and context-aware learning (DARMA-CAL) is proposed for HSI classification. In this work, we introduce the ARMA filter instead of a spectral filter to apply to GNNs. The ARMA filter can better capture the global graph structure and is more robust to noise. More importantly, the ARMA filter can simplify calculations compared with the spectral filter. In addition, we show that the ARMA filter can be approximated by a recursive method. Furthermore, we propose a dense structure, which not only implements the ARMA filter in the structure, but is also locality-preserving. Finally, we design a layerwise context-aware learning mechanism to extract the useful local information generated by each layer of the dense ARMA network. The experimental results on three real HSI datasets show that DARMA-CAL outperforms the compared state-of-the-art methods.**

***Index Terms*— Autoregressive moving average (ARMA) filter, graph convolution neural network, hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL images (HSIs) acquired by remote sensors contain abundant spatial and spectral information of ground objects simultaneously, which enables us to accurately recognize targets with different materials [1]. Compared with multispectral image [2] or RGB (red, green, and blue) analysis, HSI analysis has great power to recognize the objects of interest more effectively. As a result, HSI classification, which aims to categorize each image pixel into a certain label, has attracted wide attention in many fields such as agricultural monitoring, military detection, and disaster prevention and control [3]. However, high spectral mixing between materials [4], complex noise effects and spectral variability [5], and labeled training sample deficiency [6] bring great difficulties to HSI classification information extraction. In the past several decades, researchers have successfully designed various handcrafted [7]–[11] methods for HSI classification. Traditional machine learning algorithms [12]–[14], such as the support vector machine (SVM) [15], random forest, and K-nearest-neighbor (KNN) [14], have achieved great success in HSI classification. However, traditional handcrafted-based methods are quite empirical and depend heavily on professional expertise [16].

The successful application of deep learning techniques [17] led to their utilization in HSI classification. One advantage is that deep learning techniques can automatically learn effective feature representations for a problem domain, thereby avoiding complicated handcrafted feature engineering [18]. Chen *et al.* [19] first attempted to utilize stacked autoencoder networks to extract high-level features from spectral dimensionally reduced HSI yielded by principal component analysis (PCA) for HSI classification. Afterward, Li *et al.* [20] employed a deep belief network machine and restricted Boltzmann to HSI feature extraction.

Meanwhile, generative recurrent neural networks (RNNs) [21], [22] and adversarial networks (GANs) [23] [24] were applied in hyperspectral image classification. Among these deep learning methods, the convolutional neural network (CNN) demonstrated outstanding performance in HSI classification because it can extract more spectral features of HSIs [25]. Hu *et al.* [26] first applied CNN to hyperspectral classification and obtained better performance than SVM. However, the original CNN method only extracted the spectral information and did not make full use of the space-spectral information of HSIs.

To alleviate these deficiencies, many new types of networks have emerged. For example, in [27], residual blocks were employed to improve the representation ability of CNNs, which are capable of extracting spectral signatures

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and spatial contexts of HSIs to improve the classification rate. Lan *et al.* [28] proposed a novel dilated CNN for road segmentation from remote-sensing images. Although spectral–spatial features can be automatically extracted by CNN-based methods, the fixed convolution kernel is the bottleneck of CNN application in different regions and an HSI, which may result in undesirable misclassifications. To address this problem, people try to use different convolution kernels according to different features. Feng *et al.* [6] adopted different sizes and locations of spatial windows according to sample-specific distribution, which brings new ideas to the classification of CNN-based methods.

Comparatively, GCNs [29] are a heavily researched topic. Different from CNNs, GCNs conduct semi-supervised learning on graph-structured data (graph-based representations of molecules and social network data [30]–[32]). Extensive research has been conducted on HSI classification. Qin *et al.* [33] took advantage of the spectral–spatial information for HSI classification by adopting GCN, which greatly improved classification accuracy. However, the spectral–spatial GCN (S^2GCN) method treats each pixel as a graph node and brings a significant amount of computation. Hong *et al.* [34] proposed taking advantage of GCN and CNN by combining them, which opened up new ideas for HSI classification. In [35], superpixel segmentation was employed on an HSI, and a multiscale GCN was adopted to extract multiscale graph features. Recently, more graph filters, for example, autoregressive moving average (ARMA) filter [36], are applied to graph networks instead of spectral filters. Nevertheless, the following common shortcomings still exist in GCNs.

1) GCNs require high computational cost (resulting from the spectral filter), which is a significant bottleneck in the HSI classification task.
2) In addition, GCNs (using a spectral filter) cannot effectively suppress noise, and the classification effect for classes with similar spectral features is not ideal.
3) Finally, the traditional GCNs fail to effectively preserve the local features of each convolution layer, which leads to oversmoothing (the representation of each node tends to be congruent) as the number of convolution layers becomes large. For this reason, GCNs cannot be designed too deeply, which constrains the networks from extracting high-level features of the graph.

In response to previous problems and further boost the performance of HSI classification, in this study, we propose a semi-supervised locality-preserving dense graph neural network (GNN) with ARMA filters and a context-aware learning (DARMA-CAL) method. In this work, we introduce the ARMA filter instead of the spectral filter to apply to GNNs. The ARMA filter can capture the global graph structure better and is more robust to noise. More importantly, the ARMA filter can simplify calculations compared with the spectral filter. In addition, we prove that the ARMA filter can be approximated by a recursive method. Furthermore, we propose a dense structure with an ARMA filter (DARMA) that not only implements the ARMA filter in structure, but is also locality-preserving. Finally, we design a layerwise context-aware learning mechanism to extract the useful local information generated by each layer of the dense ARMA network. In summary, the main contributions of this article are as follows.

1) We introduce the ARMA filer into the GCN for HSI classification. The ARMA filter is more robust to noise and can simplify calculations.
2) We propose a novel dense structure. The structure can not only implement the ARMA filter, but can also solve the oversmoothing problem of GCN.
3) We design a layerwise context-aware learning mechanism to extract useful local information that can automatically extract useful feature information according to the classification target to improve the adaptability of the proposed end-to-end trainable model.

II. Review of GNN Filter

In this part, some preliminaries of the GNN filter are provided by reviewing basic notation and definitions. We assume a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$, where $\mathcal{V}$ is the vertex set with $|\mathcal{V}| = N$. $\mathcal{E}$ presents the edge set, the adjacency matrix of the graph is $\boldsymbol{A} \in \mathbb{R}^{N\times N}$, and a_{ij} denotes the weight of the edge between vertex i and vertex j.

A. Graph Spectral Filter

The Laplacian matrix $\boldsymbol{L}$ for a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$ is calculated as follows:

$$\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{A} \tag{1}$$

where $\boldsymbol{D}$ is the degree matrix of the graph.

The corresponding symmetrically normalized Laplacian $\boldsymbol{L}_{\text{sym}}$ is as follows:

$$\boldsymbol{L}_{\text{sym}} = \boldsymbol{I}_N - \boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{D}^{-\frac{1}{2}} \tag{2}$$

where $\boldsymbol{I}$ is an identity matrix.

Given functions f and g, the convolution can be calculated as follows:

$$f(t) \star g(t) \triangleq \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \tag{3}$$

where $\star$ is the convolution operator, and τ denotes the shifting distance.

Theorem 1: The Fourier transform of the convolution of two functions f and G is the product of their corresponding Fourier transforms. This can be expressed as follows:

$$\mathcal{F}[f(t) \star g(t)] = \mathcal{F}[f(t)] \cdot \mathcal{F}[g(t)] \tag{4}$$

where $\cdot$ and $\mathcal{F}$ denote the pointwise multiplication and Fourier transform, respectively.

Given the Fourier transform theorem 1, the spectral decomposition $\boldsymbol{L}_{\text{sym}}$ can be written as follows:

$$\begin{aligned}\boldsymbol{L}_{\text{sym}} &= \boldsymbol{U}\ \boldsymbol{diag}[\lambda_1, \ldots, \lambda_n]\boldsymbol{U}^T \\ &= \sum_{n=1}^{N} \lambda_n \boldsymbol{u}_n \boldsymbol{u}_n^T\end{aligned} \tag{5}$$

where $\boldsymbol{U}$ is the matrix of eigenvectors of $\boldsymbol{L}_{\text{sym}}$, that is, $UU^T = E$; λ_n denotes the eigenvalue; and $\boldsymbol{u}_n$ represents the eigenvector.

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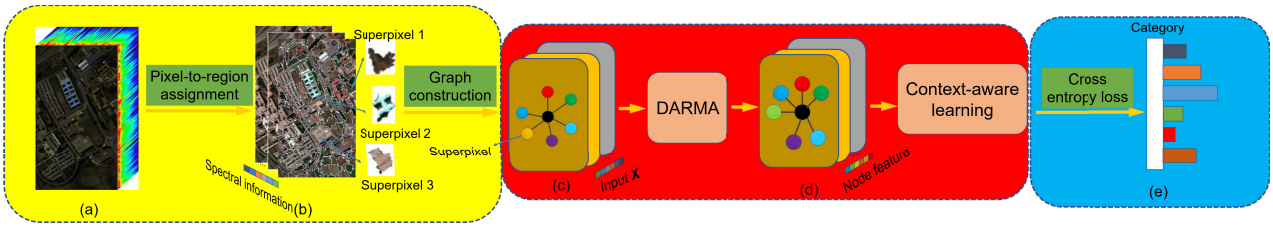


Fig. 1. Overview of DARMA-CAL network.

A graph filter can be defined as an operator that modifies the signal $\boldsymbol{X}$ on the eigenvector basis of $\boldsymbol{L}_{\text{sym}}$ according to the frequency response h applied to each eigenvalue λ_n. We can express the filtered graph signal $\bar{\boldsymbol{X}}$ as follows:

$$\begin{aligned}\bar{\boldsymbol{X}} &= \sum_{n=1}^{N} h(\lambda_n)\boldsymbol{u}_n\boldsymbol{u}_n^T\boldsymbol{X} \\ &= \boldsymbol{U}\ \text{diag}\ [h(\lambda_1),\ldots,h(\lambda_N)]\boldsymbol{U}^T\boldsymbol{X}. \end{aligned} \tag{6}$$

Spectral filters are widely used with GNNs. However, the spectral filter is not localized, the full projection on the eigenvector will produce a path of infinite length, and the filter will act on each node of the entire graph map rather than the graph node neighborhood. In addition, the main drawback of spectral filtering is its computational cost since the spectral decomposition of $\boldsymbol{L}$ is expensive in computation, and $\boldsymbol{U}$ in (6) is full.

B. Graph ARMA Filter

The ARMA model is one of the most widely used autoregressive models in time-series prediction. ARMA assumes the future value as a linear combination of past errors and past values [36]. The variable X_t^{A} at time t can be calculated as follows:

$$X_t^A = \sum_{i=1}^{\alpha}\phi_i X_{t-i} - \sum_{j=1}^{\beta}\psi_j\epsilon_{t-j} + \epsilon_t \tag{7}$$

where ϵ_{t-j} represents the random error at time $t-j$, ϕ_i and ψ_j denote the coefficients, X_{t-i} is the past value at time $t-i$, and β and α are moving average polynomials and autoregressive, respectively. If $\psi_j = 0$, then (7) is converted to an autoregressive (AR) model. If $\phi_i = 0$, then (6) can be converted to the moving average (MA) model.

Assume a dynamic process on the graph $\boldsymbol{X}^{(0)}, \boldsymbol{X}^{(1)}, \boldsymbol{X}^{(2)}, \ldots, \boldsymbol{X}^{(k)}, \ldots, \boldsymbol{X}^{(0)} = \boldsymbol{X}$ and $\boldsymbol{X}^{(k+1)} = \boldsymbol{L}\boldsymbol{X}^{(k)}$. The filtered graph signal $\bar{\boldsymbol{X}}$ is represented by the MA term as follows:

$$\bar{\boldsymbol{X}} = \psi_0\boldsymbol{X}^{(0)} + \psi_1\boldsymbol{X}^{(1)} + \cdots + \psi_{K-1}\boldsymbol{X}^{(K-1)}. \tag{8}$$

The filtering operation performed by the AR term can be expressed as follows:

$$\bar{\boldsymbol{X}} = \phi_0\bar{\boldsymbol{X}}^{(0)} + \phi_1\bar{\boldsymbol{X}}^{(1)} + \cdots + \phi_K\bar{\boldsymbol{X}}^{(K)}. \tag{9}$$

By expanding (8) and (9) explicitly, (7) can be expressed as follows:

$$\left(\boldsymbol{I} + \sum_{k=1}^{K}\phi_k\boldsymbol{L}^k\right)\bar{\boldsymbol{X}} = \left(\sum_{k=0}^{K-1}\psi_k\boldsymbol{L}^k\right)\boldsymbol{X}. \tag{10}$$

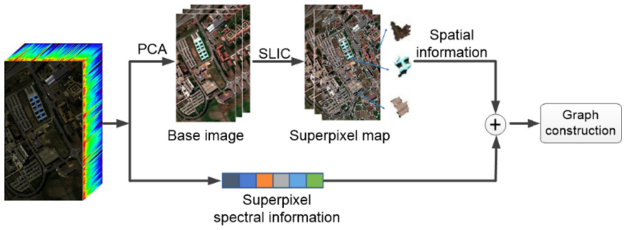


Fig. 2. Outline of HSI preprocessing. Mean spectral feature of the contained pixels are adopted as superpixel feature vector.

Then, (10) can be converted into the filtering relationship in the graph node space as follows:

$$\bar{\boldsymbol{X}} = \left(\boldsymbol{I} + \sum_{k=1}^{K}\phi_k\boldsymbol{L}^k\right)^{-1}\left(\sum_{k=0}^{K-1}\psi_k\boldsymbol{L}^k\right)\boldsymbol{X}. \tag{11}$$

The ARMA filter can capture the global graph structure better and is more robust to noise since the influence of the convolution process on the result is considered and a noise term is added to the model. However, the calculation of (11) is still complicated. This will be addressed in Section III-B.

III. Methodology

The details of DARMA-CAL will be described in this section for HSI classification (see Fig. 1). First, the input HSI [Fig. 1(a)] is segmented into a small compact superpixel map [Fig. 1(b)]. Next, DARMA is constructed to refine the superpixel graph [Fig. 1(c)]. Finally, the classification result is produced [Fig. 1(e)] by presenting contextual information via context-aware learning. In the following, the critical operations in the proposed DARMA-CAL are presented by pixel-to-region assignment and graph construction (Section III-A), expounding the ARMA graph convolutional layer (Section III-B), elaborating the neighborhood aggregation with dense connections (Section III-C), describing the attention-based global layerwise aggregation (Section III-D), and explaining the DARMA-CAL manipulation (Section III-E).

A. Pixel-to-Region Assignment and Graph Construction

We will describe HSI preprocessing in detail in this section. Fig. 2 shows an outline of the HSI preprocessing. In this article, we treat superpixels as graph nodes to improve the computational efficiency of the proposed method.

To construct a graph, we assume a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A})$. The adjacency matrix $\boldsymbol{A}_{ij} \in \mathbb{R}^{N\times N}$ of graph $\mathcal{G}$ can be

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expressed as follows:

$$\boldsymbol{A}_{ij} = \begin{cases} e^{-\gamma \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}, & \text{if } \boldsymbol{x}_i \in N(\boldsymbol{x}_j) \text{ or } \boldsymbol{x}_j \in N(\boldsymbol{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ denote two graph nodes i and j (i.e., superpixels in the paper), $N(\boldsymbol{x}_j)$ denotes the neighbors set of $\boldsymbol{x}_j$, and γ represents an empirical set to 0.2 in our proposed method.

However, there still exists a computational problem since a significant number of pixels are contained in the HSI spatial dimension. The more nodes there are for subsequent convolution and classification, the more computing ability is required, which presents a challenge to the practicality of the HSI classification algorithm. To address this problem, image segmentation technology is adopted to divide the HSI into small blocks, and the pixels contained in the small blocks have a strong spatial–spectral correlation.

Assume an HSI dataset $\{x_1, x_2, \ldots, x_m\}$ with B-bands and m pixels. The superpixel map can be mathematically expressed as follows:

$$\text{HSI} = \cup_{i=1}^{N} S_i, \quad S_i \cap S_j = \emptyset, \ i \neq j; \ i, j = 1, 2, \ldots, N \tag{13}$$

where N is the total number of superpixels and S_i denotes a superpixel containing n_i pixels.

It is impossible to segment HSI into superpixels by using classical segmentation algorithms directly since the classical segmentation method was originally designed for segmenting RGB images, and the dimension of its workspace cannot exceed three dimensions. Therefore, to split the HSI into superpixels, it is necessary to perform a spectral dimensionality reduction on the HSI in advance. In the proposed method, the popular unsupervised PCA [37] method is used to reduce the spectral dimension of the HSI, and the first principal component is taken to generate the base image, which contains abundant original HSI information. Then, we adopt the simple linear iterative clustering (SLIC) [38] method to segment the entire base HSI. Finally, we take the mean spectral features of the contained pixels in superpixels as the input vector of the proposed method, which can suppress noise influence.

B. ARMA Graph Convolutional Layer

Compared with a polynomial filter, the ARMA graph filter allows for capturing more global structures and longer dependencies [36]. However, matrix inversion in (11) produces a dense matrix and is an example of computational complexity. More importantly, the ARMA filter cannot implement GNN sparsely. To circumvent this problem, the implementation of the ARMA filter can be changed. In this article, we propose to approximate the ARMA filter by a recursive method.

Proof: We have the following proof by referring to [36]. We present the first-order recursion ARMA_1 filter as follows:

$$\bar{\boldsymbol{X}}^{(t+1)} = a\boldsymbol{M}\bar{\boldsymbol{X}}^{(t)} + b\boldsymbol{X} \tag{14}$$

where $\boldsymbol{M} = (1/2)(\lambda_{\max} - \lambda_{\min})\boldsymbol{I} - \boldsymbol{L}$, and $\lambda_{\max}$ and $\lambda_{\min}$ are the max and min eigenvalues of $\boldsymbol{L}$, respectively. a and b are correlation coefficients.

From the convergence of (14), the frequency response of an ARMA_1 filter can be expressed as follows:

$$\bar{\boldsymbol{X}} = \lim_{t\to\infty}\left[(a\boldsymbol{M})^t \bar{\boldsymbol{X}}^{(0)} + b\sum_{i=0}^{\tau}(a\boldsymbol{M})^i \boldsymbol{X}\right]. \tag{15}$$

In (15) for $|a| < 1, \mu_m \in [-1, 1]$, $(a\boldsymbol{M})^t$ goes to zero as $t \to \infty$ for any $\bar{\boldsymbol{X}}^{(0)}$. When $t \to \infty$, according to the geometric series convergence theorem, the geometric series $b\sum_{i=0}^{\tau}(a\boldsymbol{M})^i \boldsymbol{X}$ can converge to a matrix, namely

$$\lim_{t\to\infty}\left(b\sum_{i=0}^{\tau}(a\boldsymbol{M})^i \boldsymbol{X}\right) = b/(1 - a\mu_m) \tag{16}$$

where $\mu_m = (\lambda_{\max} - \lambda_{\min})/2 - \lambda_m$, and μ_m and λ_m denote the mth eigenvalues of $\boldsymbol{M}$ and $\boldsymbol{L}$, respectively. Then, we can write (15) in the following form:

$$h_{\text{ARMA}_1}(\mu_m) = \frac{b}{1 - a\mu_m}. \tag{17}$$

The analytical form of the ARMA_K filter in (11) can be restored by summing the Kth ARMA_1 filter, and the final filtering operation can be expressed as follows:

$$\bar{\boldsymbol{X}} = \sum_{k=1}^{K}\sum_{m=1}^{M}\frac{b_k}{1 - a_k\mu_m}\boldsymbol{u_m}\boldsymbol{u_m^T}\boldsymbol{X} \tag{18}$$

where $h_{\text{ARMA}_k}(\mu_m) = \sum_{k=1}^{K}(b_k)/(1 - a_k\mu_m)$. Equations (11) and (18) are mathematically equivalent, which proves that the ARMA filter can be approximated by a recursive method.

The graph convolutional operation of ARMA_1 can be calculated as follows:

$$\bar{\boldsymbol{X}}^{(t+1)} = \sigma\left(\tilde{L}\bar{\boldsymbol{X}}^{(t)}\boldsymbol{W} + \boldsymbol{X}\boldsymbol{V}\right) \tag{19}$$

where $\boldsymbol{V}$ and $\boldsymbol{W}$ denote trainable parameters, $\bar{\boldsymbol{X}}$ presents the node features, σ is the logistic ReLu function, and $\tilde{\boldsymbol{L}} = \boldsymbol{M}$ denotes the modified Laplacian by setting $\lambda_{\max} = 2$ and $\lambda_{\min} = 0$, which can compensate for small deviations introduced by the trainable parameters $\boldsymbol{W}$ and $\boldsymbol{V}$. Each ARMA_1 layer extracts local substructure information by aggregating node information in local neighborhoods. In addition, the ARMA_1 layer is the basic convolutional layer in our proposed DARMA model. Then, the output of the ARMA_l convolutional layer is expressed as follows:

$$\text{ARMA}_l = H\left(\bar{\mathbf{X}}^{(l-1)}\right) \tag{20}$$

where $\bar{\mathbf{X}}^{(l-1)}$ is the output of the ARMA_{l-1} convolutional layer, and H denotes the graph convolution operation of ARMA_1 as (19).

C. Neighborhood Aggregation With Dense Connections

As described above, we can approximate the ARMA filter by a recursive method. However, the recursion process cannot save intermediate information for subsequent processing. In addition, the graphical representation capabilities of traditional GCNs are limited to globalization and excessive refinement and neglect locality preservation, which leads to an oversmoothing problem [39]. The implementation structure

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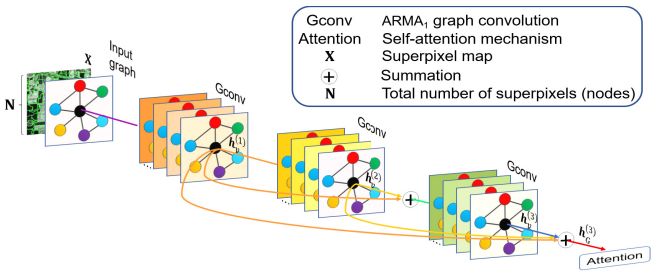


Fig. 3. Illustration of three-layer locality-preserving DARMA for HSI classification. Each layer consists of an ARMA_1 graph convolutional layer. The circles and lines represent the superpixel (graph node) and edges, different colors of the nodes represent different land-cover types. Different colors of "Gconv" represent different convolution layers.

of [36] is complex from the perspective of practical application, and it is not suitable for HSI classification. To address these issues, we directly connect every hidden convolutional layer to all subsequent convolutional layers. The resulting architecture is similar to DenseNets, which are widely adopted in computer vision problems [40]. Fig. 3 demonstrates the layout of the resulting three-layer locality-preserving the DARMA network schematically. Consequently, the lth-layer output receives the features of all preceding graph layers, and the lth-layer output $\boldsymbol{h}_v^{(l)}$ can be computed as follows:

$$\boldsymbol{h}_v^{(l)} = H\left(\boldsymbol{h}_G^{(l-1)}\right) \tag{21}$$

where $\boldsymbol{h}_v^{(l-1)}$ is the output of the $l-1$ layer, and H denotes the graph convolution operation of ARMA_1, as shown in (19). As shown in Fig. 3, we can express the lth-layer output of DARMA $\boldsymbol{h}_G^{(l)}$ as follows:

$$\boldsymbol{h}_G^{(l)} = H\left(\boldsymbol{h}_G^{(l-1)}\right) + \sum_{k=0}^{l-1} \boldsymbol{h}_v^{(k)}. \tag{22}$$

When $3 < l < 7$, (22) can also be transformed into the following form:

$$\boldsymbol{h}_G^{(l)} = \left[\text{ARMA}_1 + (l-1)\text{ARMA}_2 + \left[(k-5)(l-2) + \sum_{k=0}^{l-3} k\right] \times \text{ARMA}_k \cdots + (l-1)\text{ARMA}_{l-1} + \text{ARMA}_l\right]\boldsymbol{X} \tag{23}$$

where l is the convolution layer of the proposed network, k denotes kth ARMA_k filter (local feature). From (22) and (23), we conclude that DARMA realizes the function of ARMA graph convolution and preserves the local information of each layer of convolution.

D. Attention-Based Global Layerwise Aggregation

With the introduction of DARMA, each convolution layer can retain all feature information of the previous convolution layer. However, as the depths of the network increase, the output layer contains increasing feature information, which causes difficulty in extracting useful information. And different features are required for different targets. To obtain global contextual features in the graph, a graph attention mechanism is employed in the proposed network to extract different association degrees between different layers. The relationship between any two layers in the output is calculated via a graph attention mechanism to adapt the specific task. Given the layerwise graph-level representations, that is, $\{\boldsymbol{h}_v^{(1)}, \ldots, \boldsymbol{h}_v^{(l)}\}$, a layer-focused attention mechanism is adopted to compute graph-level output. The attention score α_l for layer l ($\sum_{l=1}^{L} \alpha_l = 1$) can be computed as follows:

$$\alpha_l = \text{softmax}\left(\boldsymbol{W}_2 \cdot \sigma\left(\boldsymbol{W}_1 \cdot \boldsymbol{h}_v^{(l)}\right)\right) \tag{24}$$

where $\boldsymbol{W}_1, \boldsymbol{W}_2 \in \mathbb{R}^{d_o \times d_o}$ are the trainable weight matrices; $\sigma(\cdot)$ denotes a nonlinear layer, that is, LeakyReLU$(\cdot)$; and a softmax function is adopted for normalization. Then, the final aggregated $\boldsymbol{h}_G^{(\text{final})}$ is expressed as a weighted average of the previous layer features, which can be calculated as follows:

$$\boldsymbol{h}_G^{(\text{final})} = \sigma\left(\sum_{k=0}^{K} \alpha_k \left(\boldsymbol{W}_1 \cdot \boldsymbol{h}_v^{(k)}\right)\right) \tag{25}$$

where k denotes the network layer of DARMA and a_k is the learned attention weight in (24). $\boldsymbol{h}_G^{(\text{final})}$ contains rich feature information for different HSI classifications, which enhances the adaptability of the model to different hyperspectral image classification tasks.

E. DARMA-CAL Manipulation

Equations (12) and (13) show the HSI preprocessing, which uses the input of the subsequent networks. Then, a DARMA network is introduced to extract the graph features. Finally, context-aware learning is conducted on the network, and the output of DARMA-CAL is $\boldsymbol{h}_G^{(\text{final})}$. In the proposed DARMA-CAL, a cross-entropy error is adopted to penalize the difference between the DARMA-CAL output and the label examples, namely

$$\boldsymbol{L} = -\sum_{z \in \mathbf{y_G}} \sum_{f=1}^{C} \boldsymbol{Y}_{zf} \ln \boldsymbol{h}_{Gzf}^{(\text{final})} \tag{26}$$

where $\mathbf{y_G}$ is the labeled example set, $\boldsymbol{Y}_{zf}$ presents the label matrix, and C denotes the number of classes. Similar to [41], Adam gradient descent is adopted to update the parameters for DARMA-CAL. The implementation details of our DARMA-CAL are shown in Algorithm 1.

IV. Experimental

In this part, extensive experiments are conducted to validate the performances of the proposed DARMA-CAL method. First, DARMA-CAL is compared with six state-of-the-art methods on three real HSI datasets. Then, the impact of hyperparameters on DARMA-CAL performance is analyzed. Furthermore, the ablative effects of the dense structure and context-aware learning are investigated. Finally, the running time of our model is demonstrated.

A. Dataset

Three real benchmark datasets, that is, Pavia University, Salinas, and Houston 2013, are employed to evaluate the performances of the DARMA-CAL method both quantitatively and qualitatively.

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Algorithm 1 Mechanism of DARMA-CAL

Input: HSI; learning rate = 0.001; number of epoch T;
1: Segment the original HSI into superpixel by PCA-SLIC method;
2: Superpixel spectral features are extracted, and graph is constructed by Eq. (12) and Eq. (13);
3:// DARMA-CAL model is trained
4: **for** t = 1 to T **do**
5: Extract the graph features by Eq. (22);
6: Bach normalization, dropout and ReLu;
7: Assign different weights to features of different layers by Eq. (24);
8: Perform context-aware learning by Eq. (25);
9: Error term is calculated by Eq. (26), and weight matrices are updated by utilizing Adam;
10: **end for**
11: Predict pixel label based on trained network.
Output: Each pixel label.

TABLE I
AMOUNTS OF TRAINING AND TESTING PIXELS IN PAVIA UNIVERSITY DATASET

Class ID	Class name	Training	Testing
1	Asphalt	30	6601
2	Meadows	30	18619
3	Gravel	30	2069
4	Trees	30	3034
5	Painted-metal-sheets	30	1315
6	Bare-soil	30	4999
7	Bitumen	30	1300
8	Self-blocking-bricks	30	3652
9	Shadows	30	917

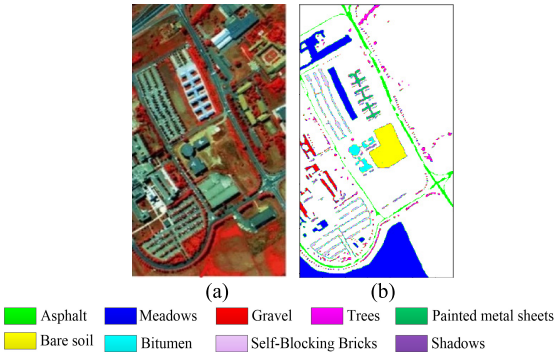


Fig. 4. Pavia University. (a) False color image. (b) Ground truth map.

1) Pavia University: The first dataset, Pavia University, is part of the hyperspectral data of an image of Pavia City, Italy, which was acquired by an airborne reflection optics spectral imaging system (ROSIS) in 2003. The dataset contains 103 bands and 610 × 340 pixels. The whole map can be categorized into nine kinds of features, which are shown in Fig. 4. Table I lists the amount of data used for training and testing.

2) Salinas: The second dataset, Salinas, was collected by the 224-band AVIRIS sensor over the region of Salinas Valley, CA, USA. Salinas contains 512 lines by 217 samples. After discarding 20 bands that cannot be reflected by water, 204 bands remain. These pixels are divided into 16 categories,

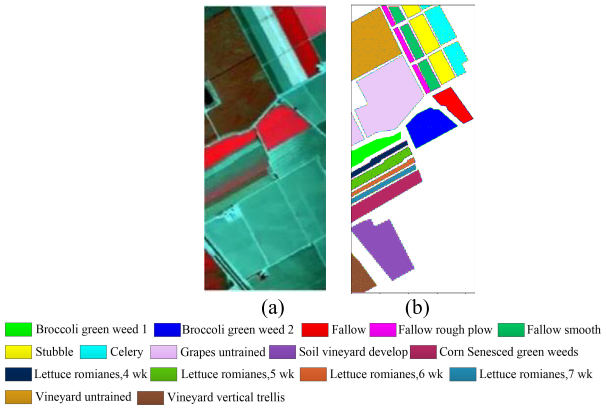


Fig. 5. Salinas. (a) False color image. (b) Ground truth map.

TABLE II
AMOUNTS OF TRAINING AND TESTING PIXELS IN SALINAS DATASET

Class ID	Class mane	Training	Testing
1	Broccoli-green-weed 1	30	1979
2	Broccoli-green-weed 2	30	3696
3	Fallow	30	1946
4	Fallow-rough-plow	30	1364
5	Fallow-smooth	30	2648
6	Stubble	30	3929
7	Celery	30	3549
8	Grapes-untrained	30	11241
9	Soil-vineyard-develop	30	6137
10	Corn-Senesced-green-weeds	30	3248
11	Lettuce-romianes-4wk	30	1038
12	Lettuce-romianes-5wk	30	1897
13	Lettuce-romianes-6wk	30	886
14	Lettuce-romianes-7wk	30	1040
15	Vineyard-untrained	30	7238
16	Vineyard-vertical-trellis	30	1777

TABLE III
AMOUNTS OF TRAINING AND TESTING PIXELS IN HOUSTON 2013 DATASET

Class ID	Class name	Training	Testing
1	Healthy-grass	30	1221
2	Stressed-grass	30	1224
3	Synthetic-grass	30	667
4	Tress	30	1214
5	Soil	30	1212
6	Water	30	295
7	Residential	30	1238
8	Commercial	30	1214
9	Road	30	1222
10	Highway	30	1197
11	Railway	30	1205
12	Parking-Lot-1	30	1203
13	Parking-Lot-2	30	439
14	Tennis-Court	30	398
15	Running-Track	30	630

which are demonstrated in Fig. 5. The amount of data for training and testing is shown in Table II.

3) Houston 2013: The third dataset, Houston 2013, contains 349 × 1905 pixels and 144 spectral bands. The dataset contains 15 categories, which are shown in Fig. 6. The amount of data for training and testing is shown in Table III.

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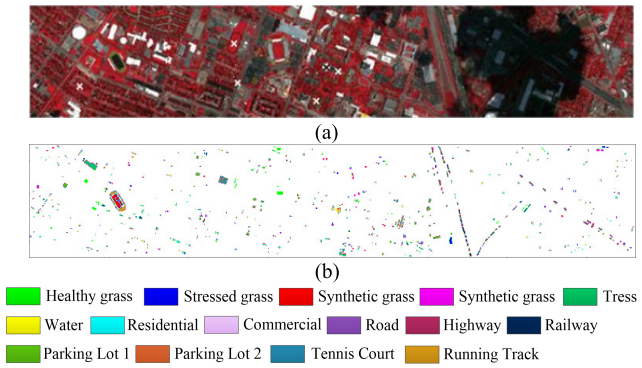


Fig. 6. Houston 2013. (a) False color image. (b) Ground truth map.

TABLE IV
HYPERPARAMETER SETTINGS FOR DIFFERENT DATASETS

Dataset	N	*lr*	*L*	*T*
University of Pavia	20000	0.001	3	500
Salinas	600	0.001	3	1000
Houston University	35000	0.001	3	500

B. Experiment Setup

In the experiments, DARMA-CAL is conducted via PyTorch with the Adam [42] optimizer. The hyperparameter selection in our DARMA-CAL, including segmentation scale N, learning rate lr, number of DARMA layers L, and number of iterations T, is shown in Table IV.

To quantitatively compare the six methods on three real HSI datasets, the per-class accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are adopted to evaluate the proposed performance.

For the aforementioned three HSI datasets, 30 labeled pixels are randomly selected in each class for training, and the remaining pixels are adopted for testing. Furthermore, 90% of the training set is randomly utilized to update the network parameters during training, and the remaining 10% of samples are adopted to tune the parameters.

To validate the performance of DARMA-CAL, we compare the proposed network with two CNN methods, that is, multi-scale dense network (MSDN) [43] and convolutional RNN (CRNN) [21], and two GCN methods, that is, spectral–spatial GCN (S^2GCN) [33] and multiscale dynamic GCN (MDGCN) [35]. Furthermore, we employed two traditional machine learning methods, that is, joint collaborative representation and SVM with decision fusion (JSDF) [13] and multiband compact texture units (MBCTU) [11]. The architectural details of the proposed network are listed in Table V. All methods were performed ten times, and the average accuracy and standard deviation of these ten independent implementations are shown.

C. Comparisons With Other Methods

In this section, we evaluate the classification performance of DARMA-CAL quantitatively and qualitatively by comparing the proposed method with the aforementioned six state-of-the-art methods.

1) Results on Pavia University Dataset: The classification results are quantitatively reported in Table VI obtained by

TABLE V
ARCHITECTURAL DETAILS OF PROPOSED NETWORK

Module	Details	
Pixel-to-region assignment	PCA SLIC	Spectral information of each region (input)
Graph construction	Calculate adjacency matrix $\boldsymbol{A}$ of graph	
Original graph processing	ARMA dense network (see Fig. 3) ARMA_1 layer=3, ReLu and BN between two layers	
Context-Aware learning	Layerwise graph attention layer BN	
Output	Cross-entropy (target categories)	

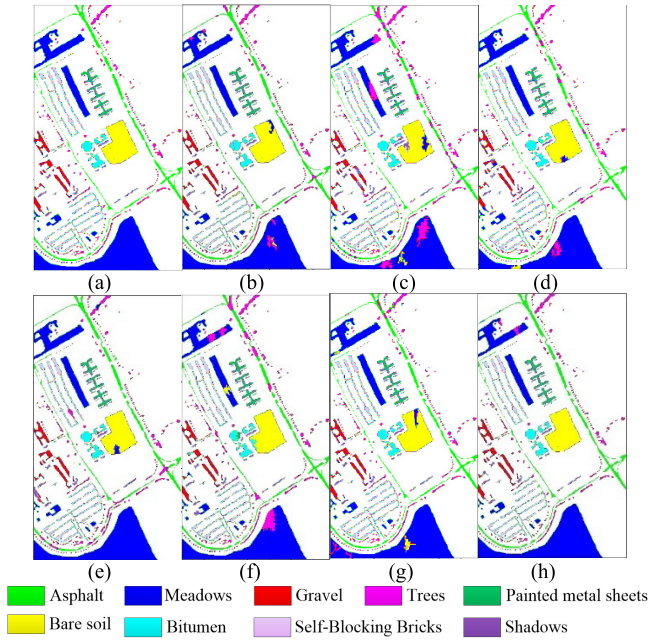


Fig. 7. Classification maps on Pavia University dataset. (a) Ground-truth map. (b) MSDN. (c) CRNN. (d) S^2GCN. (e) MDGCN. (f) MBCUT. (g) JSDF. (h) DARMA-CAL.

the aforementioned methods in terms of OA, AA, and κ on the Pavia University dataset, and we highlight the highest index of each row in bold. From the table, we observe that the proposed DARMA-CAL outperforms the compared method in OA, AA, and κ. This denotes the advancement of the model and the theoretical design. It is also notable that the GCNs achieved good classification results, especially the MDGCN method. Although the MDGCN model achieved a good result, its classification accuracy in $C4$ (self-blocking bricks) and $C9$ (shadows) is significantly lower than that of DARMA-CAL, which indicates that our proposed method has a good adaptation for HSI classification details.

Fig. 7 shows a visual comparison of the HSI classification results produced by the aforementioned methods above on the Pavia University dataset. A visual inspection reveals that the proposed DARMA-CAL method demonstrates fewer misclassifications than the ground-truth map. Meanwhile, due to the lack of a local feature-preserving mechanism, the detailed results produced by the compared methods contain many misclassifications.

2) Results on Salinas Dataset: The quantitative results of seven methods on the Salinas dataset are listed in Table VII. $C8$ (Grapes untrained) and $C15$ (Vineyard untrained) have

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TABLE VI
QUANTITATIVE COMPARISON FOR PAVIA UNIVERSITY, WITH BEST PERFORMANCE IN BOLD

Class No.	MSDN	CRNN	S^2GCN	MDGCN	MBCUT	JSDF	DARMA-CAL
1	91.20±1.62	79.85±4.62	92.87±3.79	93.55±0.37	87.49±3.99	82.40±4.07	**97.76±1.67**
2	94.61±0.59	82.33±3.17	87.06±4.47	**99.25**±0.23	89.11±5.58	90.76±3.74	97.67±1.32
3	87.18±2.33	89.67±3.69	87.97±4.77	92.03±0.24	86.24±4.23	86.71±4.14	**97.87±1.78**
4	90.29±3.21	91.45±2.44	90.85±0.94	83.78±1.55	90.61±3.39	92.88±2.16	**96.21±2.33**
5	95.64±0.76	94.12±1.78	**100.00±0.00**	99.47±0.09	97.18±1.18	**100.00±0.00**	96.05±2.45
6	88.89±1.28	91.37±2.11	88.69±2.64	95.26±0.50	93.25±2.93	94.30±4.55	**99.82±0.59**
7	92.48±2.36	93.67±1.97	98.88±1.08	98.92±1.04	93.49±2.47	96.62±1.37	**99.77±0.20**
8	82.67±1.34	80.18±3.29	89.97±3.28	94.99±1.33	84.14±4.78	94.69±3.74	**96.28±1.65**
9	95.17±0.57	82.34±4.86	98.89±0.53	81.03±0.49	96.57±1.22	**99.56±0.36**	96.62±0.71
OA (%)	90.90±1.49	85.46±1.75	89.74±1.70	95.68±0.22	89.43±2.14	90.82±1.30	**97.71±0.82**
AA (%)	91.12±1.56	87.22±1.82	92.80±0.47	93.15±0.28	90.90±0.89	93.10±0.65	**97.56±0.53**
Kappa	91.55±1.63	84.19±1.43	86.65±2.06	94.25±0.29	86.24±2.62	88.02±1.62	**98.24±0.76**

TABLE VII
QUANTITATIVE COMPARISON FOR SALINAS, WITH BEST PERFORMANCE IN BOLD

Class No.	MSDN	CRNN	S^2GCN	MDGCN	MBCUT	JSDF	DARMA-CAL
1	98.21±0.67	99.34±0.53	99.01±0.44	99.98±0.03	99.18±0.80	**100.00±0.00**	**100.00±0.00**
2	99.62±0.19	99.17±0.21	99.18±0.59	99.90±0.28	99.76±0.33	**100.00±0.00**	**100.00±0.00**
3	97.43±0.52	96.54±1.39	97.15±2.76	99.80±0.21	99.13±1.04	**100.00±0.00**	**100.00±0.00**
4	97.62±0.97	97.32±0.27	99.11±0.55	97.49±2.16	97.61±0.82	**99.93±0.09**	99.42±0.57
5	95.31±1.21	98.75±0.89	97.55±2.35	97.96±0.77	96.54±1.01	**99.77±0.31**	98.04±1.27
6	99.44±0.26	99.19±0.77	99.32±0.35	99.10±1.67	99.74±0.32	**100.00±0.00**	99.34±0.61
7	97.28±0.61	98.67±1.30	90.06±0.27	98.18±1.49	98.26±1.64	**99.99±0.01**	99.86±0.12
8	77.21±2.37	72.38±3.98	70.68±5.20	92.78±4.61	81.98±4.32	87.79±4.89	**95.99±2.18**
9	98.62±0.51	97.29±0.61	98.32±1.79	**100.00±0.00**	99.47±0.51	99.67±0.33	99.98±0.02
10	93.18±1.39	91.44±1.64	90.97±2.59	**98.31±1.29**	92.21±2.75	96.53±2.55	92.27±2.36
11	96.21±0.39	96.82±0.78	98.00±1.65	99.39±0.55	96.24±2.68	**99.76±0.21**	97.61±1.21
12	99.99±0.01	99.21±0.32	99.56±0.59	99.01±0.78	98.98±0.45	**100.00±0.00**	99.27±0.53
13	96.49±0.92	97.29±0.86	97.83±0.72	97.59±1.32	96.73±1.66	**100.00±0.00**	96.43±1.31
14	95.21±0.79	95.10±1.73	95.75±1.65	97.92±1.72	96.50±3.05	**98.71±0.72**	93.52±2.69
15	73.62±2.31	76.33±4.62	70.36±3.62	95.71±4.57	79.41±5.67	81.86±5.26	**96.44±0.92**
16	96.31±0.52	97.99±0.61	96.90±1.97	98.18±2.92	96.89±2.19	98.99±0.63	**99.89±0.20**
OA (%)	91.64±1.67	87.64±0.83	88.39±1.01	97.25±0.87	92.14±0.86	94.67±0.77	**97.96±0.37**
AA (%)	94.48±0.85	94.55±0.57	94.30±0.47	**98.21±0.30**	95.54±0.56	97.69±0.34	98.00±0.62
Kappa	90.62±0.89	86.72±1.01	87.10±1.12	96.94±0.96	91.25±0.95	94.06±0.85	**97.43±0.92**

similar spectral signatures, and the classification results are lower than those of the other class. However, the proposed method still achieves a good classification effect since the ARMA filter has a suppressive effect on noise. In addition, JSDF achieved better classification results in most classes. However, the performance on OA and κ of the two methods is lower than that of DARMA-CAL since there is an imbalance among different class classifications. As a visual comparison demonstrated in Fig. 8, our proposed DARMA-CAL is noticeably closer to the ground truth map [see Fig. 8(a)] than the other six competitors, which further shows the advantage of DARMA-CAL.

3) Results on Houston 2013 Dataset: As shown in Table VIII, the classification result of all investigated methods decreased compared with the other two datasets in terms of three quantitative criteria. The Houston 2013 dataset is larger, and more details need to be classified, which puts forward higher requirements for algorithm feature extraction. As can be observed, the proposed DARMA-CAL achieves the best classification result with the OA, AA, and κ coefficients, which further validates the advantages of the proposed model feature extraction.

Not surprisingly, the performance of GCN-based methods is generally superior to that of CNN-based methods owing to the ability to extract the spatial relation between samples. It is also notable that JSDF achieves higher accuracies in seven land cover classes, which shows that traditional machine learning methods have their application advantages in a given dataset if they are properly designed. Compared with the proposed DARMA-CAL, JSDF has very poor performance in $C8$ (commercial) and $C9$ (road). Fig. 9 visualizes the classification results produced by the seven investigated methods, where some critical regions are enlarged for better performance presentation. As seen from the zoomed-in regions of Fig. 9, there is an obvious misclassification in the comparison method classification map. By contrast, DARMA-CAL performs best.

D. Analysis of Performance of DARMA-CAL With Different Numbers of Labeled Samples

In this part, we investigated the performances of DARMA-CAL with different numbers of training samples (i.e., pixels). The number of training samples per class is varied from 5 to 30 with an interval of 5, and the OA performances yielded by the seven methods are shown in Fig. 10. From

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TABLE VIII
QUANTITATIVE COMPARISON FOR HOUSTON 2013, WITH BEST PERFORMANCE IN BOLD

Class No.	MSDN	CRNN	S^2GCN	MDGCN	MBCUT	JSDF	DARMA-CAL
1	92.62±1.25	82.45±2.19	96.30±3.07	93.42±4.25	92.86±3.83	**97.41±1.21**	90.52±3.62
2	95.28±0.61	84.12±3.64	98.57±1.47	93.67±3.60	92.18±2.79	**99.84±0.25**	94.26±4.19
3	92.33±0.79	91.56±1.07	98.88±0.43	98.12±1.09	97.42±1.19	99.88±0.22	**100.00±0.00**
4	93.46±1.35	91.29±3.78	97.68±2.89	95.58±1.85	90.96±1.98	**98.22±2.80**	82.31±5.31
5	99.21±0.09	98.81±0.62	97.66±1.12	99.00±1.30	97.17±1.29	**100.00±0.00**	99.92±0.21
6	88.92±1.33	94.83±2.19	96.84±1.17	93.28±6.08	91.78±3.22	**99.32±1.09**	99.00±0.74
7	85.21±0.92	86.42±3.42	83.48±5.89	87.68±4.41	82.88±3.81	91.32±4.91	**92.54±2.39**
8	79.64±3.66	53.05±8.71	76.15±4.37	80.45±6.12	71.85±5.64	68.82±6.16	**84.34±2.76**
9	81.21±2.39	84.04±4.63	82.17±1.78	**89.64±2.26**	81.94±4.25	69.47±8.56	87.40±3.20
10	78.62±3.72	45.14±8.77	86.85±8.32	90.06±6.41	87.31±5.08	85.63±9.32	**91.48±1.84**
11	86.42±1.51	61.85±9.39	88.57±5.06	86.73±3.22	77.41±6.46	94.51±3.82	**98.24±0.96**
12	81.92±1.37	84.40±2.93	78.64±4.79	**89.44±5.69**	86.35±5.85	84.33±5.33	81.22±4.17
13	83.27±0.68	84.14±3.12	75.62±6.93	92.78±4.45	85.58±5.35	**98.10±1.28**	93.08±6.41
14	97.77±0.68	96.03±0.73	99.45±0.44	99.43±0.97	96.85±1.85	**100.00±0.00**	**100.00±0.11**
15	97.65±0.24	93.45±2.41	98.03±1.07	96.27±1.72	92.27±3.32	**99.86±0.36**	95.98±4.08
OA (%)	88.49±0.96	82.10±1.21	89.31±1.00	91.40±0.92	87.07±1.12	90.51±0.95	**93.04±0.87**
AA (%)	88.90±1.37	79.21±1.02	90.33±1.06	92.37±0.89	89.32±1.08	92.46±0.75	**92.68±0.69**
Kappa	88.27±1.65	77.61±1.19	88.44±1.08	90.70±1.00	86.01±1.21	89.74±1.03	**92.36±0.92**

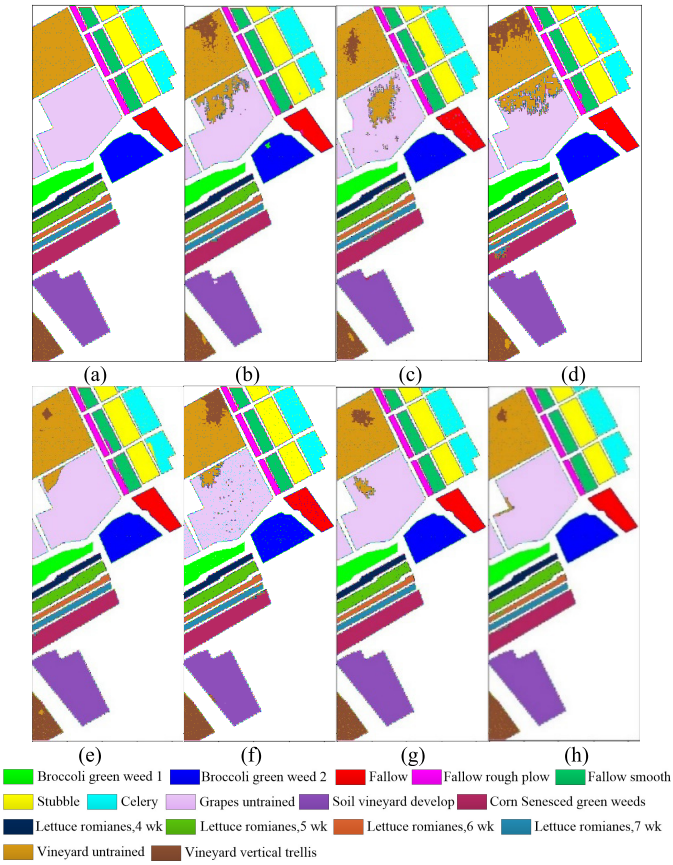


Fig. 8. Classification maps on Salinas dataset. (a) Ground-truth map. (b) MSDN. (c) CRNN. (d) S^2GCN. (e) MDGCN. (f) MBCUT. (g) JSDF. (h) DARMA-CAL.

the figure, we find that the classification performances of the seven algorithms improve significantly as the labeled examples increase because more labeled training dates are attributed to a higher classification accuracy rate. Meanwhile, the proposed DARMA-CAL generally outperforms S^2GCN, MDGCN, and all other contrast algorithms since DARMA-CAL allows for preservation of the local information of each convolutional layer and automatic learning of global contextual features. It is also worth mentioning that the OA of the proposed MARMA-CAL is stable as the number of labeled examples changes, which shows that our proposed model has better adaptability to the classification tasks and the number of training labels. The ARMA kernel can effectively suppress noise and enhance classification performance.

E. Hyperparameter Impact

In the experiment, the sensitivity of the classification performance to different hyperparameter settings was evaluated in detail. The hyperparameters to be retuned include the number of DARMA layers L, segmentation scale N, learning rate lr, and number of iterations T. Herein, a grid search strategy is adopted to find the optimal setting. The four hyperparameters are divided into two groups, and OA is reported as the change in each group of hyperparameters. The impact of the number of DARMA layers L and segmentation scale N on the three data sets are revealed in Fig. 11, and the parametric sensitivity of the learning rate lr and number of iterations T is demonstrated in Fig. 12.

From Fig. 11, we observe that the best result is reached when the number of convolution layers is 3, and the classification results do not improve with an increase in the number of convolution layers. Increasing the number of convolution layers can extract more detailed features but requires more computation and reduces the classification efficiency. Hence, the number of DARMA layers L is set to 3 in the proposed method. For N, the classification accuracy improves with an increase in the number of segmentations because a larger N results in a finer segmentation. However, the classification accuracy stops improving after reaching a certain segmentation value. Meanwhile, the computational complexity of the model also increases with an increase in the number of segmentations. In our proposed method, the segment scales S are 20 000, 600, and 35 000 for the Pavia University, Salinas, and Houston 2013 datasets, respectively.

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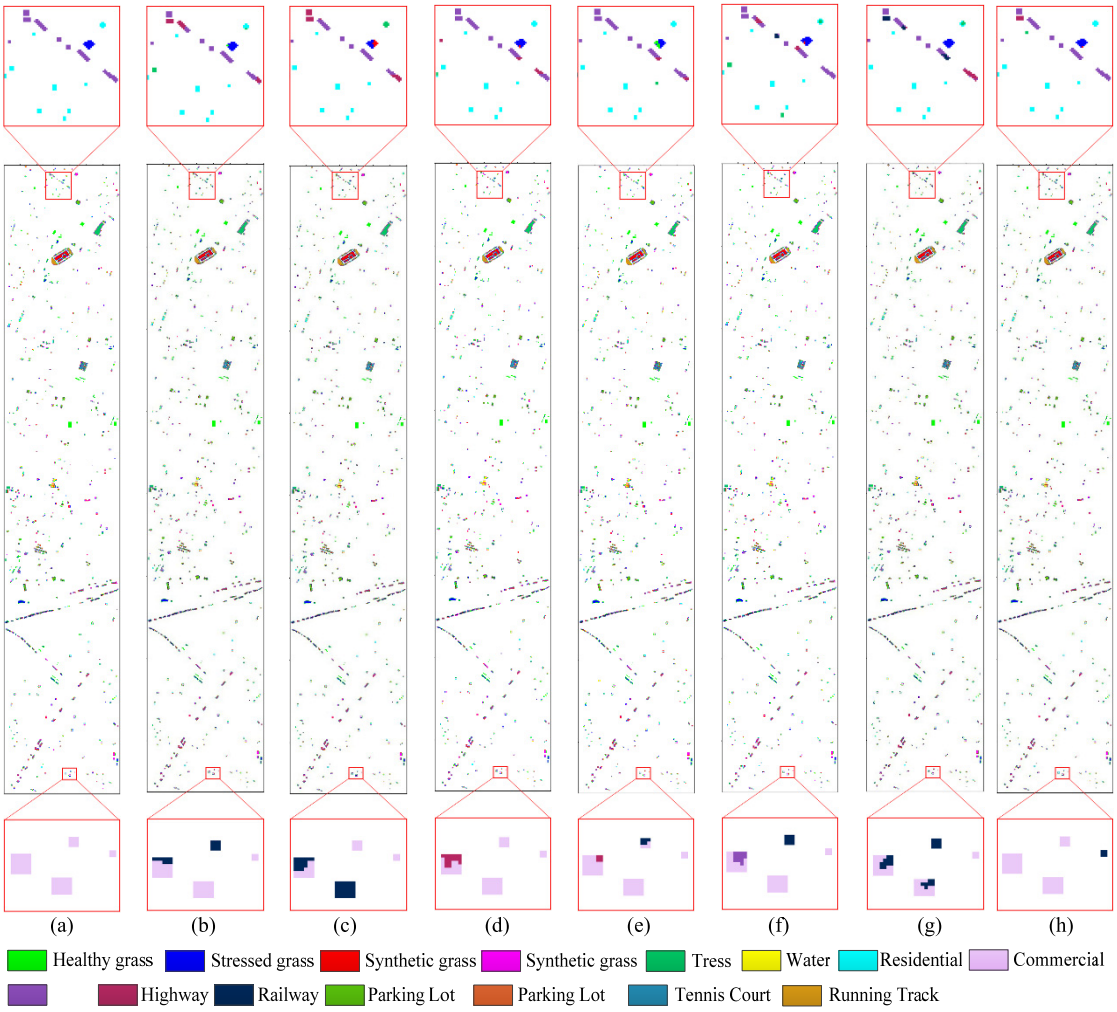


Fig. 9. Classification maps on Houston dataset. (a) Ground truth map. (b) MSDN. (c) CRNN. (d) S^2GCN. (e) MDGCN. (f) MBCUT. (g) JSDF. (h) DARMA-CAL.

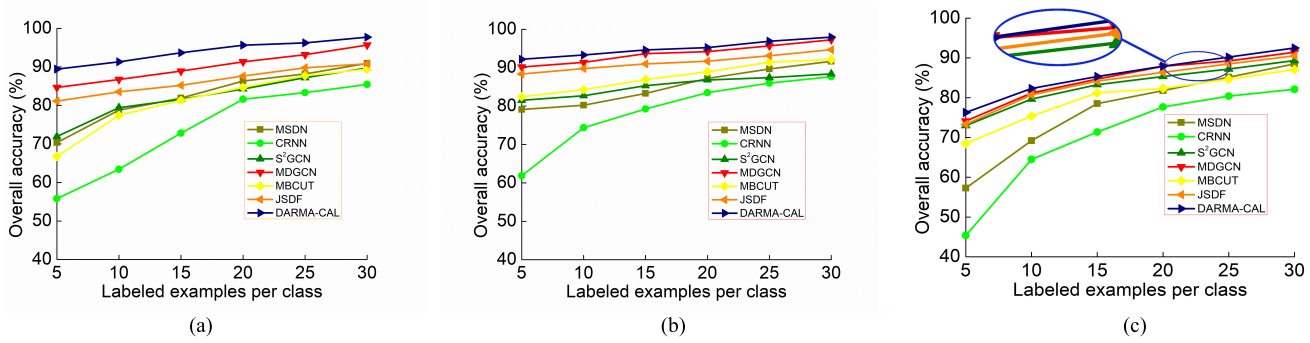


Fig. 10. Overall accuracies yielded by different methods. (a) Pavia University dataset. (b) Salinas dataset. (c) Houston 2013 dataset.

The parametric sensitivity of lr and T is presented in Fig. 12. From the results, we conclude that adopting a large lr usually cannot obtain the best results and that the results are unstable in practice. Hence, it is important to choose an appropriate learning rate. We set lr $= 0.001$, which considers both learning efficiency and classification accuracy. In addition, it is critical to utilize an appropriate T to achieve satisfactory performance. In our proposed method, the selection of T is related not only to the learning rate, but also to the number of segmentations. The best results are reached when $T = 500$, 1000, and 500 for the University of Pavia, Salinas, and Houston 2013 datasets, respectively.

F. Ablation Study

In this part, the ablative effect of DARMA-CAL is investigated. For the sake of comparison, we adopt "ARMA," "ARMA-A," and "DARMA" to represent the reduced model by removing the dense structure and the attention mechanism, the dense structure, and the attention mechanism framework, respectively. The comparative results on the three datasets

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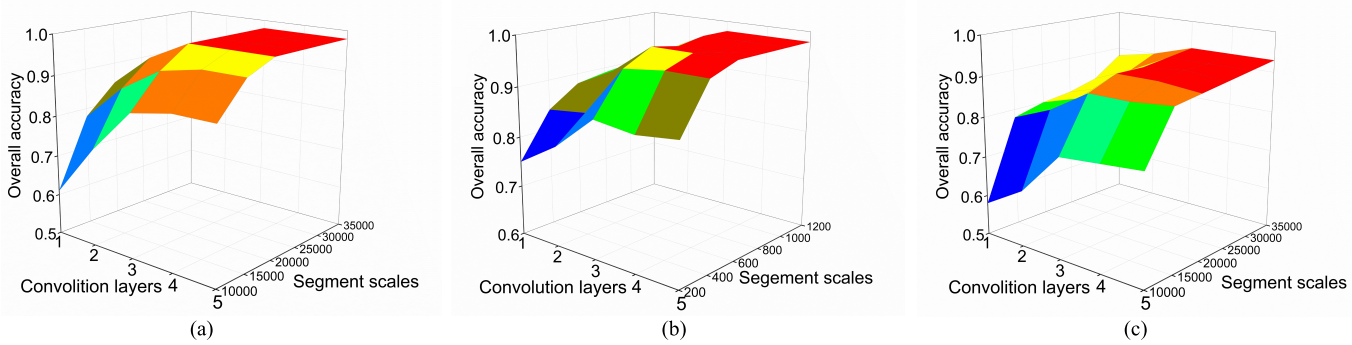


Fig. 11. Parametric sensitivity of L and N. (a) University of Pavia dataset. (b) Salinas dataset. (c) Houston 2013 dataset.

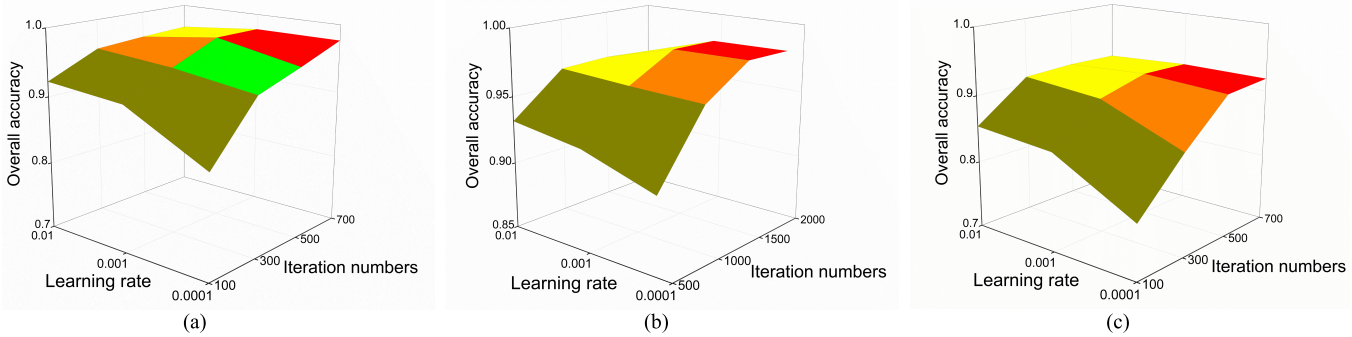


Fig. 12. Parametric sensitivity of lr and T. (a) University of Pavia dataset. (b) Salinas dataset. (c) Houston 2013 dataset.

TABLE IX
ABLATION STUDY RESULTS ON PAVIA UNIVERSITY DATASET

Methods	ARMA	ARMA-A	DARMA	DARMA-CAL
OA	91.35±0.86	95.64±0.43	93.52±0.44	**97.71±0.82**
AA	90.86±0.59	95.37±0.57	93.61±0.51	**97.56±0.53**
Kappa	91.57±0.78	95.28±0.92	94.18±0.86	**98.24±0.76**

TABLE X
ABLATION STUDY RESULTS ON SALINAS DATASET

Methods	ARMA	ARMA-A	DARMA	DARMA-CAL
OA	92.37±0.82	95.56±0.52	94.85±0.49	**97.96±0.37**
AA	92.63±0.76	95.29±0.65	95.32±0.62	**98.00±0.62**
Kappa	92.19±0.83	95.77±0.59	94.16±0.53	**97.43±0.92**

TABLE XI
ABLATION STUDY RESULTS ON HOUSTON 2013 DATASET

Methods	ARMA	ARMA-A	DARMA	DARMA-CAL
OA	89.86±0.93	91.27±0.89	90.65±0.96	**93.04±0.87**
AA	90.31±0.98	91.18±0.95	89.79±0.99	**92.68±0.69**
Kappa	89.25±1.01	90.58±0.82	90.62±0.91	**92.36±0.92**

are presented in Tables IX–XI. From the results, the dense structure and the context-aware learning mechanism play a significant role in HSI classification.

G. Training Time

To reveal the advantage of our proposed DARMA-CAL to the baselines in terms of efficiency, the running times of different deep methods are listed in Table XII, including MSDN, CRNN, S^2GCN, MDGCN, and the proposed DARMA-CAL on the three datasets. The results are reported on a server with a 3.70G Intel i9-10900K CPU and a GeForce GTX 1080Ti 11G GPU. From the results, we conclude that our proposed model is more efficient than the comparison methods, and the graph size can be significantly reduced by graph projection. Meanwhile, the utilization of segmentation operation reduces the number of computing nodes effectively. Furthermore, the ARMA filter requires less calculation, which can effectively reduce the computational cost. The comparison results demonstrate that our proposed method is effective and efficient for HSI classification.

TABLE XII
TRAINING TIME (IN SECONDS), WITH BEST IN BOLD. "PU" DENOTES UNIVERSITY OF PAVIA DATASET. "HOUSTON" DENOTES HOUSTON 2013 DATASET

Methods	MSDN	CRNN	S^2GCN	MDGCN	DARMA-CAL
PU	3027	2967	2821	2203	**1557**
Salinas	3142	2848	3026	1227	**576**
Houston	3374	2937	3534	3029	**2273**

V. CONCLUSION

In this article, a novel locality-preserving dense GNN with ARMA filters and DARMA-CAL method was proposed for HSI classification. Owing to the pixel-to-region assignment strategy, the proposed network reduced the dimensionality of HSI in the spatial dimension. Meanwhile, the strategy preserved the spectral dimension information of HSI. We innovatively utilized a dense structure to implement the ARMA filter. The structure has a better suppression of noise (due to the ARMA filter) and can preserve the local features of each convolution layer (by adapting a dense network). However, inevitably, DARMA leads to difficulty in feature extraction. To address this problem, we developed a layerwise

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context-aware learning mechanism to extract useful local information. The experimental results on three real HSI datasets showed that our proposed DARMA-CAL is advanced when compared with related state-of-the-art HSI classification methods.

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